

Software Testing & Security Checking

Project 3: Fuzz Testing

on Mobile Apps and Web Programs

Farn Wang

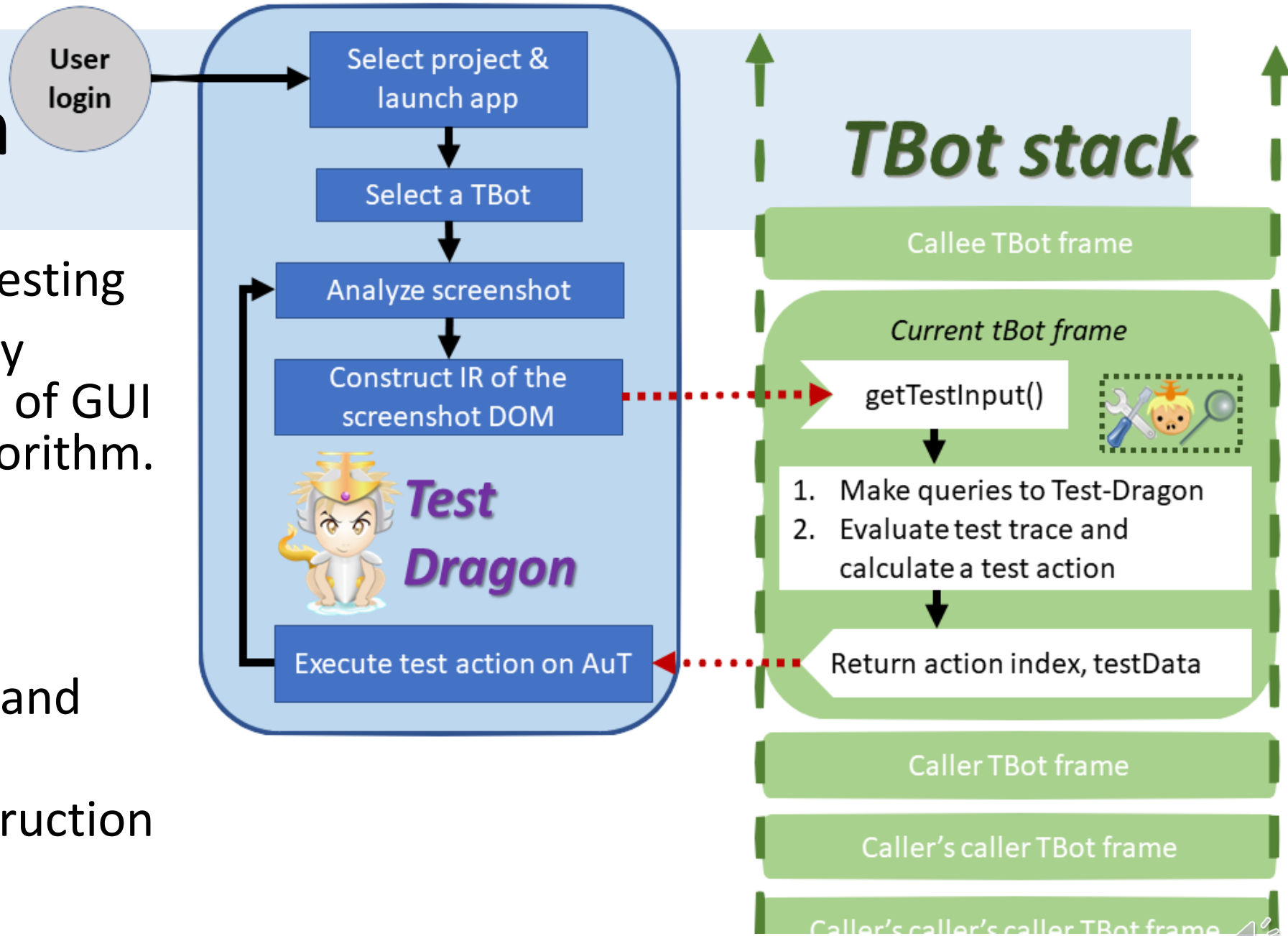
Software Testing Lab.

National Taiwan University



Test Dragon

- AI for GUI E2E-Testing
- Platform for easy implementation of GUI E2E-Testing Algorithm.
- Test Data card management & matching
- Trace recording and replay
- Automata construction



Test Dragon

- Downloading at <https://www.test-dragon.com>

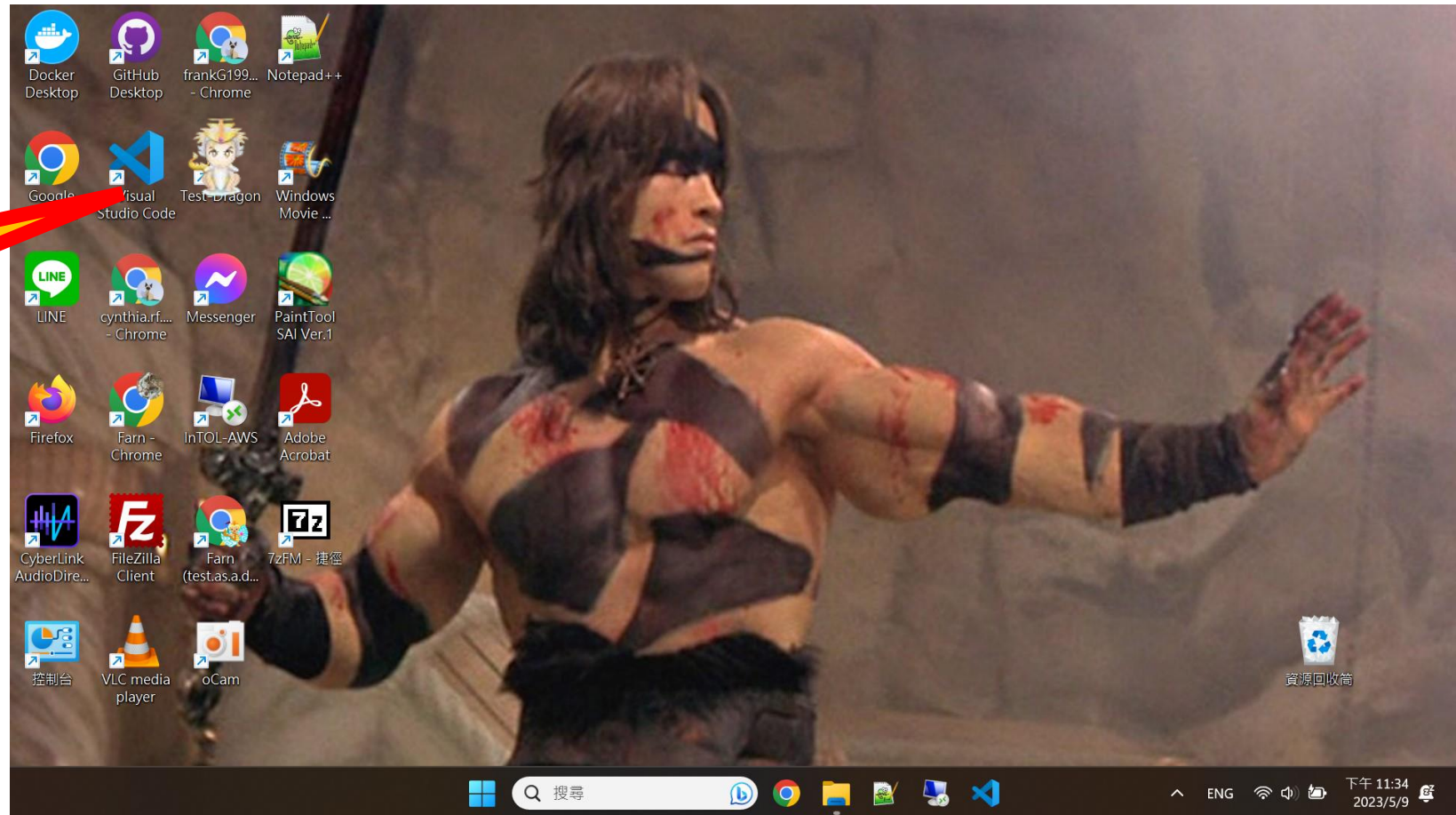
- Get an account!
- Download here!
 - ❑ pyc and python
 - ❑ Including a python interpreter.

The screenshot shows the Test-Dragon website in a web browser. The page features a header with navigation links like 'Review samples', 'Write report', and a user profile 'tupa1@gggg.invalid'. A large banner reads 'Ever since, software QA people live happily forever.' with statistics: 34 users, 106 downloads, 4230 test sessions, 100 TBot downloads, \$0 TBot sales, 576 web visits, and 3 reviews. A 'Click to download tool!' button is highlighted with a red arrow. Below the banner is a 'News!' section with a list of updates. At the bottom is the 'TBot Open Market' section with filters for 'Related categories' (e.g., 2-step verification, android, coverage) and a search bar. A table with columns 'logo', 'name', 'authors', 'sales', 'downloads', and 'reviews' is partially visible.

Test Dragon

- *Executing from desktop!*

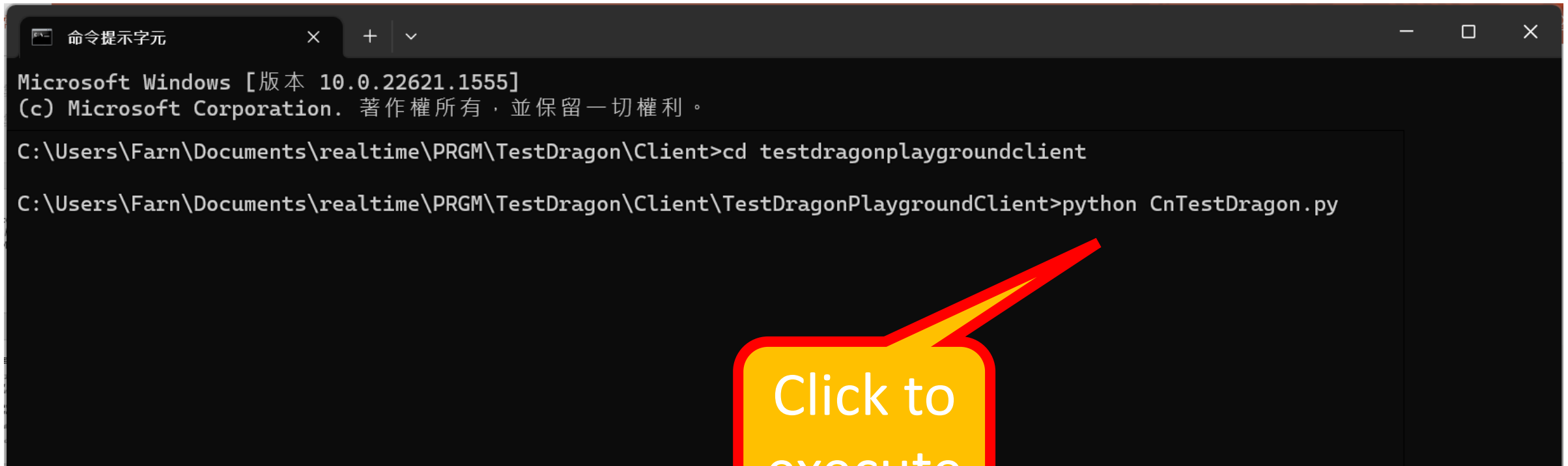
Click to
execute



Test Dragon

- *Executing from command line!*

- Must be python 3.7!



The screenshot shows a Windows Command Prompt window with a dark background. The title bar at the top reads '命令提示字元' (Command Prompt) and includes standard window controls. The main text area displays the following commands and their execution paths:

```
Microsoft Windows [版本 10.0.22621.1555]  
(c) Microsoft Corporation. 著作權所有，並保留一切權利。  
C:\Users\Farn\Documents\realtime\PRGM\TestDragon\Client>cd testdragonplaygroundclient  
C:\Users\Farn\Documents\realtime\PRGM\TestDragon\Client\TestDragonPlaygroundClient>python CnTestDragon.py
```

Click to
execute

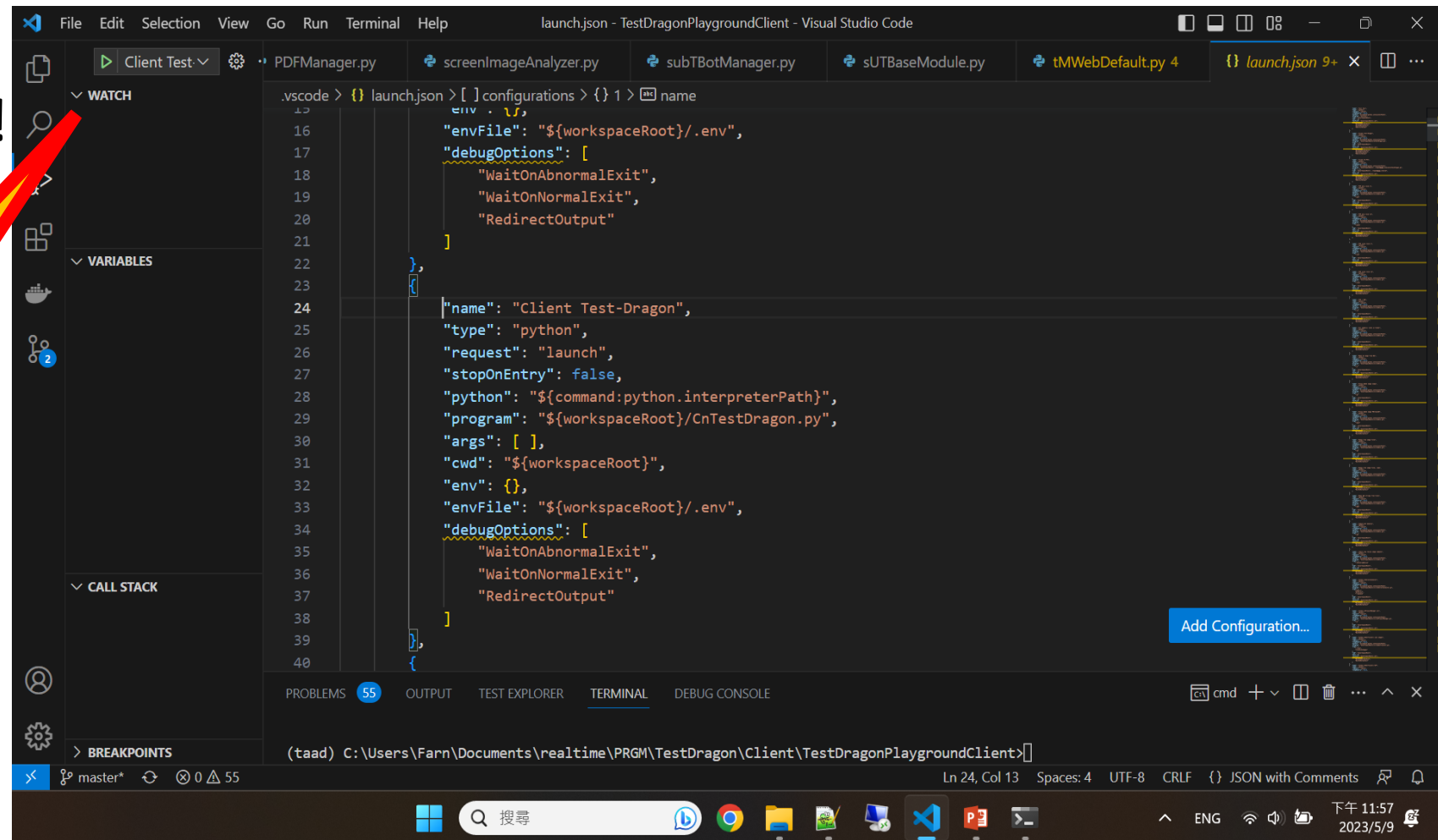


Test Dragon

- Executing from VSCode for debugging fuzzer.py and your own TBots!

❑ Must be
python 3.7!

Click to
execute



The screenshot shows the Visual Studio Code interface with a `launch.json` file open. The file contains two configuration entries. The first entry is a Python configuration named "Client Test-Dragon". The second entry is a Python configuration named "Client Test-Dragon". The first entry is highlighted with a red arrow pointing to the "Run and Debug" icon in the left sidebar. The second entry is highlighted with a red arrow pointing to the "Run and Debug" icon in the left sidebar. The first entry is highlighted with a red arrow pointing to the "Run and Debug" icon in the left sidebar. The second entry is highlighted with a red arrow pointing to the "Run and Debug" icon in the left sidebar.

```
.vscode > {} launch.json > [ ] configurations > { } 1 > name
16   "envFile": "${workspaceRoot}/.env",
17   "debugOptions": [
18     "WaitOnAbnormalExit",
19     "WaitOnNormalExit",
20     "RedirectOutput"
21   ],
22 },
23
24   {
25     "name": "Client Test-Dragon",
26     "type": "python",
27     "request": "launch",
28     "stopOnEntry": false,
29     "python": "${command:python.interpreterPath}",
30     "program": "${workspaceRoot}/CnTestDragon.py",
31     "args": [ ],
32     "cwd": "${workspaceRoot}",
33     "env": {},
34     "envFile": "${workspaceRoot}/.env",
35     "debugOptions": [
36       "WaitOnAbnormalExit",
37       "WaitOnNormalExit",
38       "RedirectOutput"
39     ],
40   },
41 }
```

PROBLEMS 55 OUTPUT TEST EXPLORER TERMINAL DEBUG CONSOLE

(taad) C:\Users\Farn\Documents\realtime\PRGM\TestDragon\Client\TestDragonPlaygroundClient>

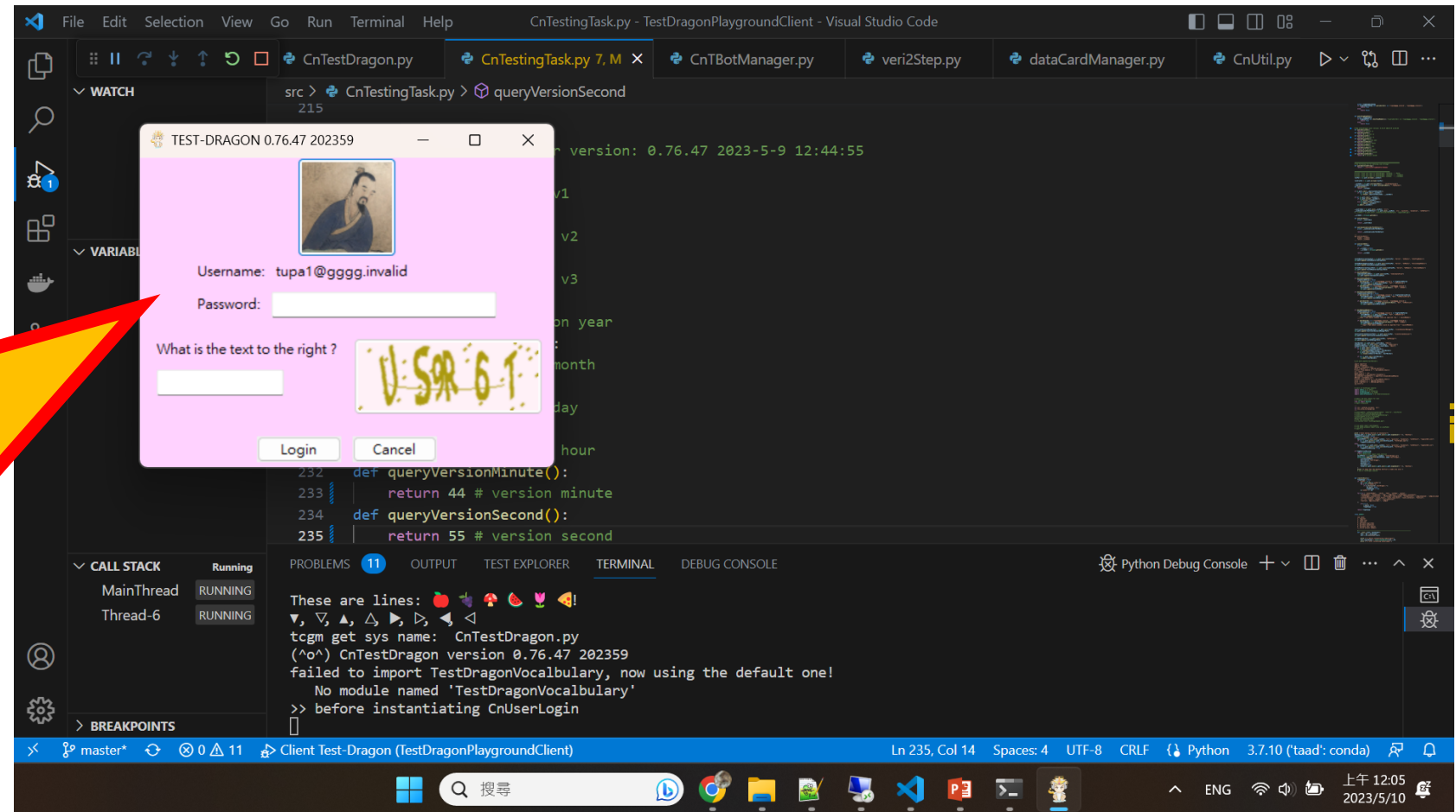
Ln 24, Col 13 Spaces: 4 UTF-8 CRLF {} JSON with Comments

Test Dragon

- After invocation, please fill in the password and captcha!

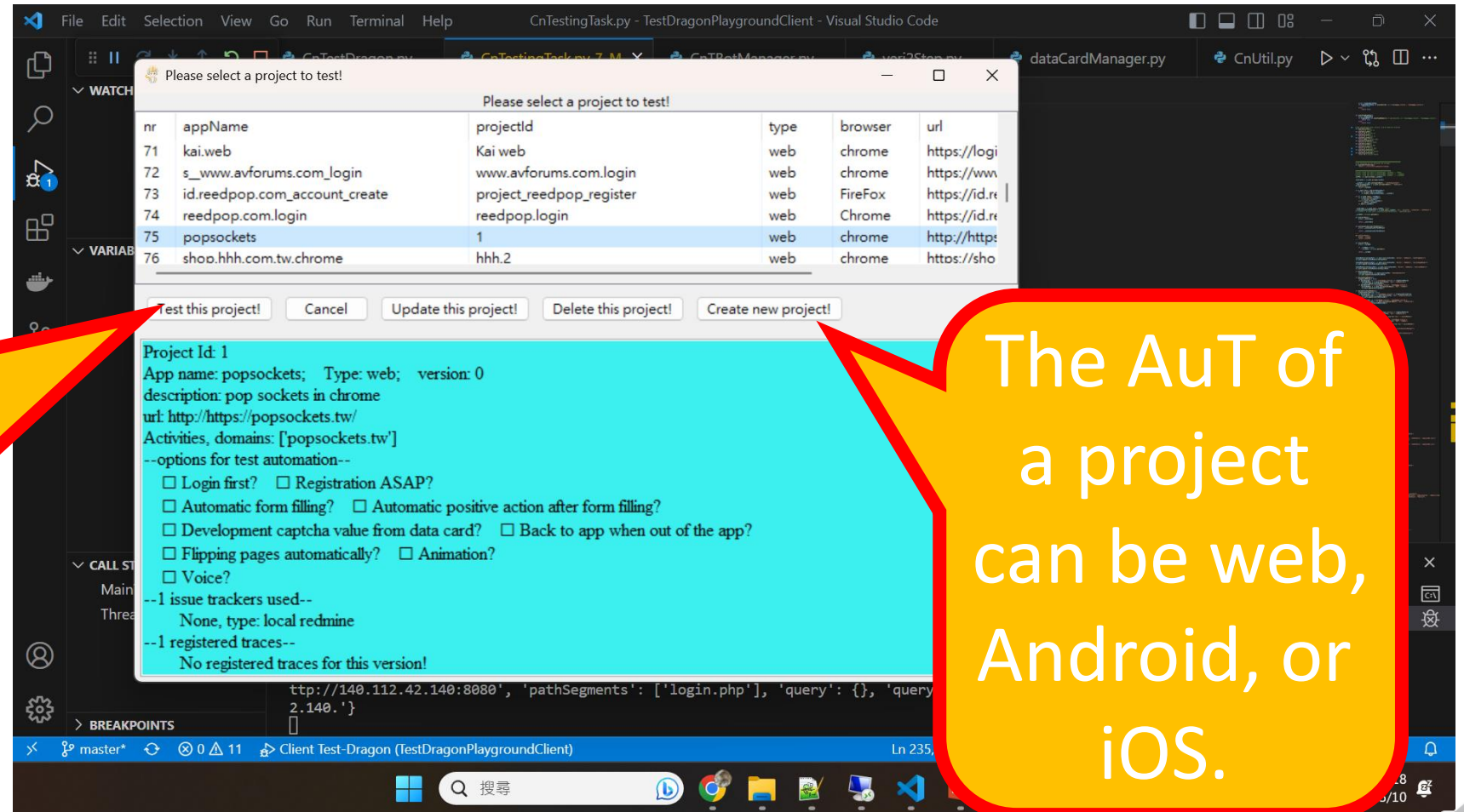
- The pyc cannot be copied to other users!

The pyc is encrypted only for you!



Test Dragon

- After logged in, select or create a project for testing!



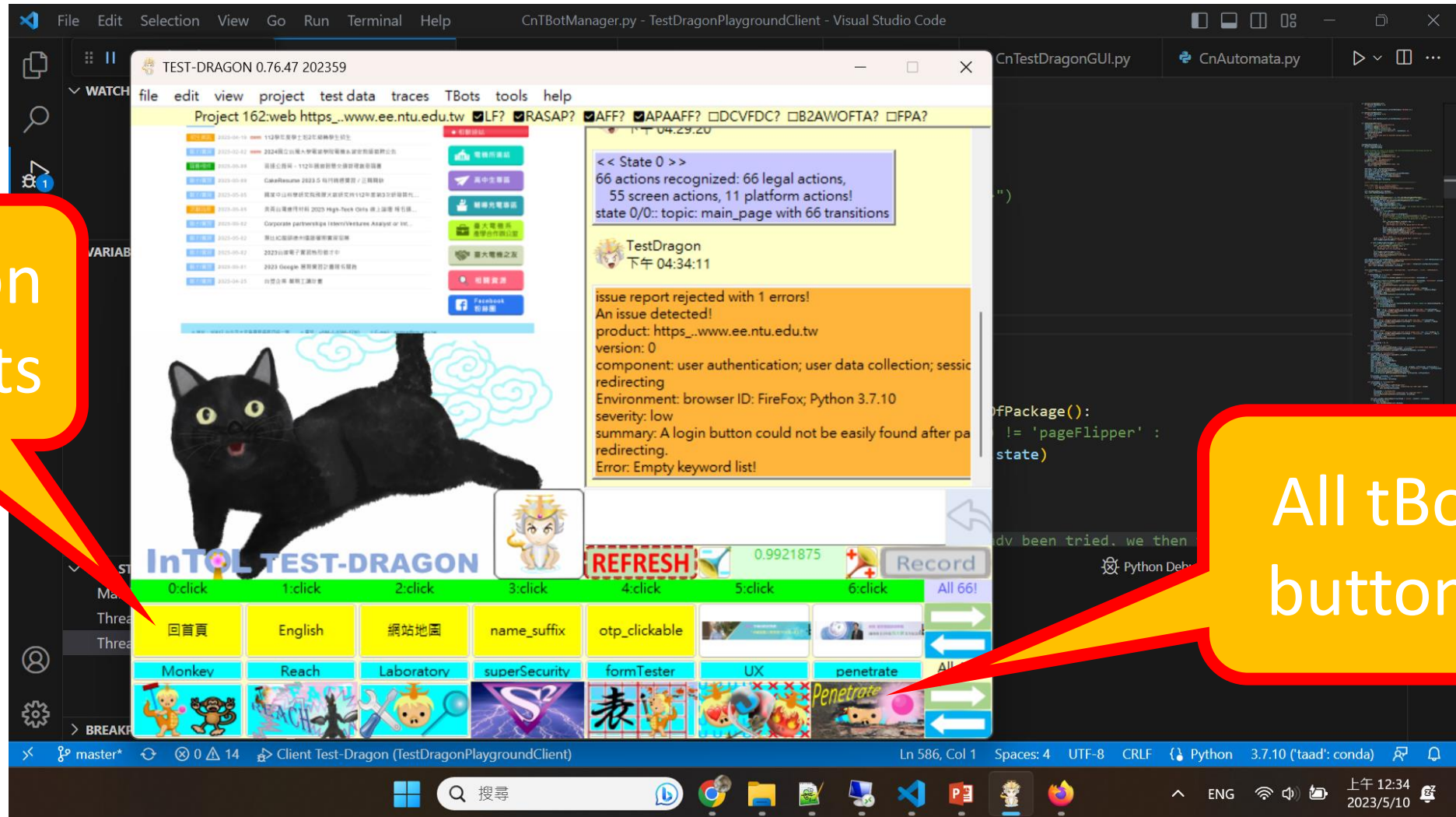
Here is selection button for project testing.

The AuT of a project can be web, Android, or iOS.

Test Dragon

- After selecting a project, now you see the full GUI of test-dragon!

All action
elements

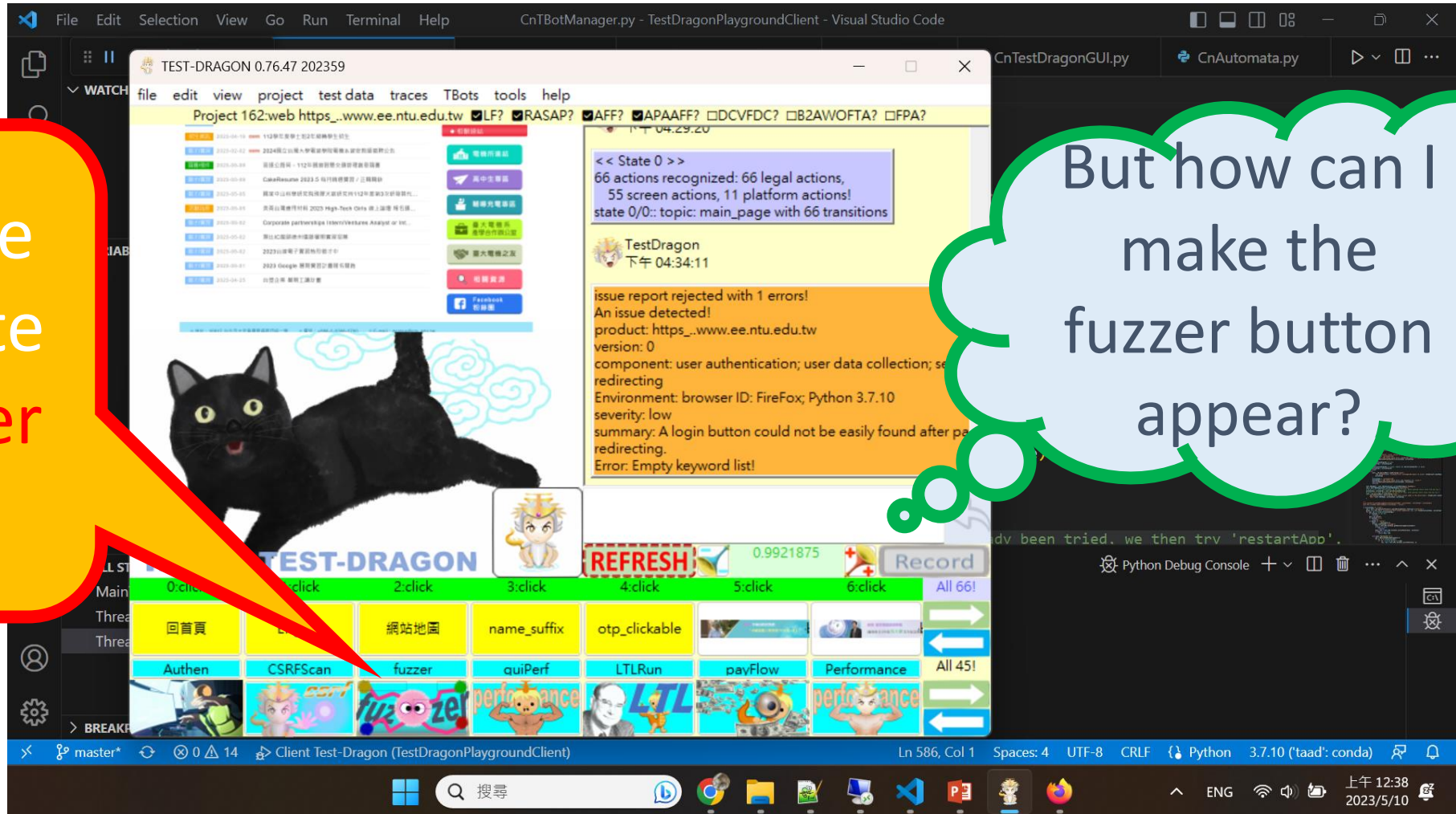


All tBot
buttons

Test Dragon

- After selecting a project, now you see the full GUI of test-dragon!

Click here
to execute
your **fuzzer**
Tbot.

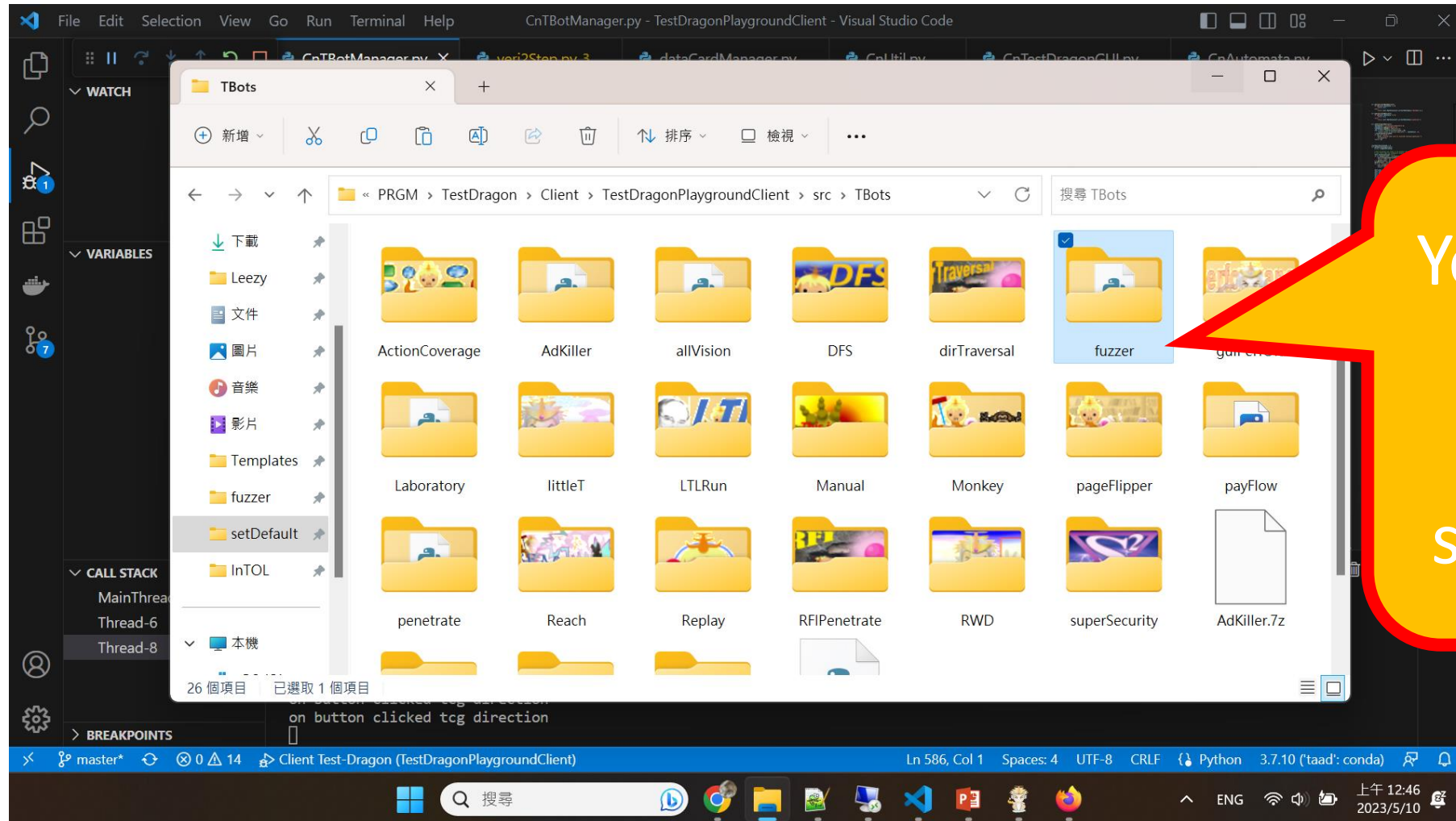


But how can I
make the
fuzzer button
appear?



Test Dragon

- How can I make the fuzzer button appear in the GUI?



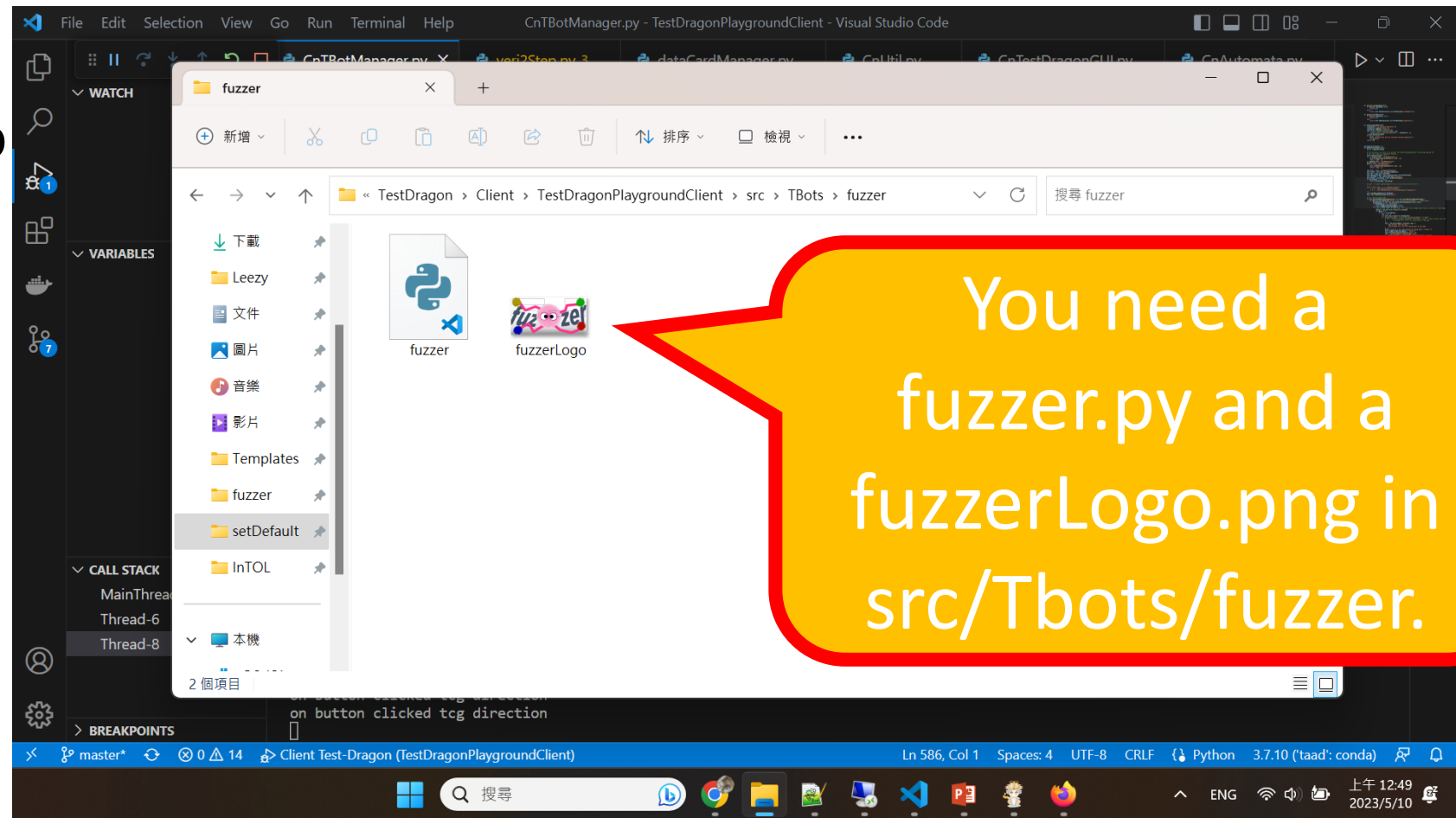
You need a
fuzzer
folder in
src/Tbots.



Test Dragon

- *How can I make the fuzzer button appear in the GUI?*

- The two files can be found in <https://github.com/farnking/fuzzer>
- You can also add issues to the github repository.



Test Dragon

- How can I get the two files for the fuzzer tBot?

- You can also find the source files in <https://www.test-dragon.com>

Click here to
see the
documents!

The screenshot shows the Test-Dragon website interface. At the top, there's a navigation bar with links like 'Review samples', 'Write report', and a user profile 'tupa1@gggg.invalid'. Below the navigation bar, the site features a cartoon dragon mascot and a large banner with the text 'Ever since, software QA people live happily forever.' followed by statistics: 34 users, 106 downloads, 4230 test sessions, 100 TBot downloads, \$0 TBot sales, 576 web visits, and 3 reviews. A 'Click to Download' button is highlighted with a red arrow pointing from a yellow callout box that says 'Click here to see the documents!'. Below the banner, there's a 'News!' section and a 'TBot Open Market' section. The 'TBot Open Market' section includes a search bar and a table with columns: logo, name, authors, sales, downloads, and reviews. The table is currently empty. The website is viewed in a browser window with the address bar showing 'test-dragon.com/home'.

Test Dragon

- How can I the two files for the fuzzer tBot?

- You can also find the source files in <https://www.test-dragon.com>

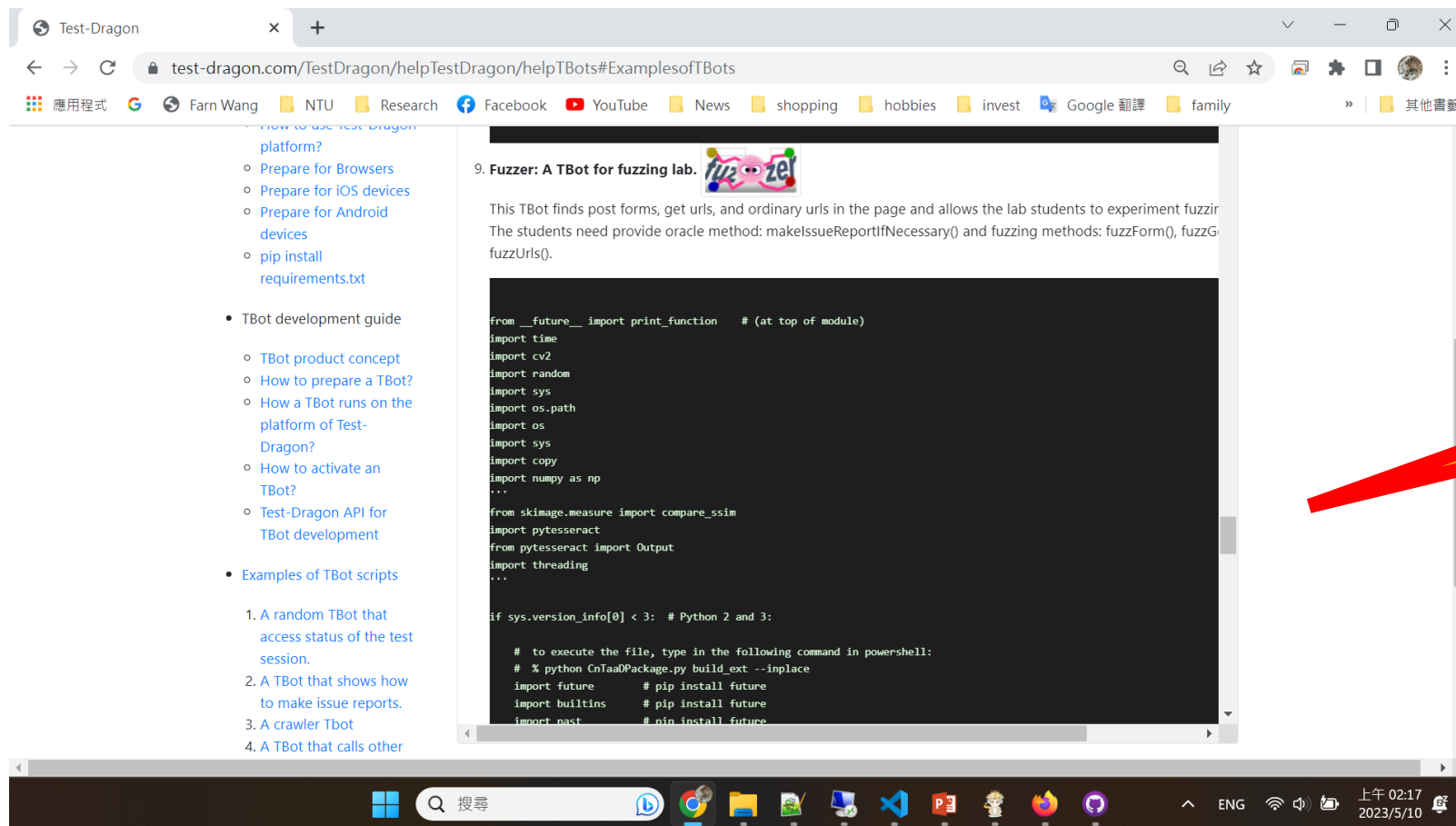
Click here to
see fuzzer.py!

The screenshot shows a web browser window with the URL `test-dragon.com/TestDragon/helpTestDragon`. The page title is "How to use Test-Dragon platform?". The sidebar on the left contains a list of links: "How to use Test-Dragon platform?", "Prepare for Browsers", "Prepare for iOS devices", "Prepare for Android devices", "pip install requirements.txt", "TBot development guide", "TBot product concept", "How a TBot runs on the platform of Test-Dragon?", "How to activate a TBot?", "How to prepare a TBot?", "Examples of TBot scripts", "Test-Dragon API for TBot development", "1. For control the platform", "2. For query and control the app under test and system under test". The main content area has three sections: "How to use Test-Dragon platform?", "Prepare for Browsers", and "Prepare for iOS devices". The "How to use Test-Dragon platform?" section says: "You can first visit the following page and use the registration link in the upper-right corner to register. <https://www.test-dragon.com>. Then you can also view your account and maintain information via the links in the upper-right corner after login." The "Prepare for Browsers" section says: "Test-Dragon web testing is built on top of selenium webdrivers. Thus for different platforms, eg. Windows, macOS, Linux, UNIX, and etc, you may need find and download your selenium webdrivers, for example [geckodriver.exe](#) for Firefox, [chromedriver.exe](#) for Chrome. You may also need to turn on your browser development mode. Sometimes, Test-Dragon may not find your webdrivers. In this case, you may need add the folder path of your webdrivers to the system path. Here may be some useful links." The "Prepare for iOS devices" section says: "There are a lot to work on for testing iOS device apps. First, the iOS development kit of Test-Dragon is built on top of Appium. Thus you need pip install Appium. Then you need install an app in your iOS device of testing so that the app can work as a proxy between the device and the XCode tool in your macOS. Since our iOS module is built on top of Appium, we also need go to get an iOS app, somewhat called webdriver agent runner-runner, as the stepstone (or backdoor, or proxy) on your iOS device desktop. This proxy says that the iOS on this device can trust your macbook for operating the device. You need go through the following procedure." The footer of the page shows the Windows taskbar with the search bar and various icons.

Test Dragon

- How can I find the two files for the fuzzer tBot?

- You can also find the source files in <https://www.test-dragon.com>



Scroll to see
fuzzer.py!

